

Algebra & Polynomials

Module 2 Test

Grade 9 • MTH1W

Time: 60 minutes • Total: 70 marks

Name: _____

Date: _____

Instructions: Show all work for full marks. Marks for each question are shown in brackets. Calculators are permitted unless your teacher states otherwise.

Section A — Knowledge & Understanding

1. Simplify: $9x - 4x$. [1]

2. Simplify: $3a + 7 - a$. [1]

3. Evaluate $4x + 1$ when $x = 2$. [1]

4. Simplify: $(2x + 5) + (3x - 2)$. [1]

5. Expand: $5(x - 3)$. [1]

6. Simplify: $(8x + 12) \div 4$. [1]

7. Simplify: $x^4 \cdot x^5$. [1]

8. State the coefficient of the term $-6y^2$. [1]

9. How many terms are in $3x^2 - 2x + 7$? **[1]**

10. Simplify: $4m + 3n - m + 2n$. **[2]**

11. Evaluate $2a + 3b$ when $a = 3$ and $b = -1$. **[2]**

12. Simplify: $(5x + 4) - (2x - 3)$. **[2]**

13. Expand: $2x(3x + 5)$. **[2]**

14. Simplify: $(9x^2 - 6x) \div 3x$. **[2]**

15. Simplify: $(4x^2)(2x^3)$. **[2]**

Section B — Application

16. Simplify: $6x^2 - 2x + 3x^2 - 5x$. **[2]**

17. Simplify: $(3a^2 - a + 2) + (a^2 + 4a - 6)$. **[2]**

18. Expand: $-3(2x - 4)$. **[2]**

19. Simplify: $(16x^2 + 8x) \div 8$. **[2]**

20. Simplify: $(20x^6) \div (5x^2)$. **[2]**

21. Evaluate $x^2 - 2x + 1$ when $x = 4$. **[2]**

22. Expand and simplify: $4(x + 2) + 3(2x - 1)$. **[3]**

23. Simplify: $(7x^2 + 3x - 2) - (2x^2 - 5x + 4)$. **[3]**

24. Simplify: $(12x^3 - 8x^2 + 4x) \div 4x$. **[3]**

Section C — Thinking

25. A triangle has perimeter $(12x + 3)$. Two sides are $(4x - 1)$ and $(5x + 5)$. Find the third side. **[4]**

26. Expand and simplify: $2x(3x - 4) - 3(x^2 - 2x)$. **[4]**

27. A rectangle has area $(6x^2 + 9x)$ and width $3x$. Find its length, then its perimeter in terms of x . (Very hard.) **[5]**
28. A rectangle has length $(3x + 1)$ and width $(x + 4)$. Write its perimeter P in simplified form, then evaluate P when $x = 5$. (Very hard.) **[5]**
29. Simplify fully: $(10x^3 - 15x^2) \div 5x + 2x(4 - x) - 3$. (Very hard.) **[6]**

Section D — Communication

30. Aisha says $3x + 3x = 3x^2$. Is she correct? Explain the difference between adding like terms and multiplying them, using this example. (Very hard.) **[4]**

— END OF EXAM —